

Experiment Log - Bilingual Prompt Classification

Project: Clarifying Prompts for Robust English-Arabic Question-Type Classification

Date: May 2026

Task: classify English and Arabic questions into NUMBER, LOCATION, PERSON, DESCRIPTION, ENTITY, or ABBREVIATION.

1. Objective

The objective was to test how prompt design affects bilingual question-type classification. The project moved from simple prompts to progressively clearer prompts, then validated the frozen final strategy across open and closed models.

2. Prompt Progression

- Pilot_03 used simple label-only prompts and reached 0.8667 English accuracy and 0.6667 Arabic accuracy.
- Pilot_04 added label definitions and improved to 0.9500 English accuracy and 0.8000 Arabic accuracy.
- Pilot_05 added one example per label and improved to 0.9667 English accuracy and 0.8667 Arabic accuracy.
- Pilot_06 added a targeted DESCRIPTION vs ENTITY disambiguation rule and improved to 0.9833 English accuracy and 0.9500 Arabic accuracy.
- Pilot_07 reused the frozen best prompt on unseen validation data and reached 1.0000 English accuracy and 0.9833 Arabic accuracy with Falcon3.

3. Final Cross-Model Validation

The frozen final prompt strategy was evaluated on Falcon3-7B-Instruct, ALLaM-7B-Instruct-preview, AceGPT-v2-8B-Chat, and Gemini-2.5-Flash. Each question was tested with eight prompt-level outputs and majority voting.

Model	Eng Acc	Ar Acc	Eng F1	Ar F1	Eng Sens	Ar Sens	Eng Cons	Ar Cons
Falcon3	1.0000	0.9833	1.0000	0.9833	0.0048	0.0105	0.9917	0.9542
ALLaM	0.9833	1.0000	0.9833	1.0000	0.0059	0.0059	0.9833	0.9833
AceGPT	0.9167	0.8833	0.9087	0.8618	0.0385	0.0313	0.8787	0.8824
Gemini	0.9833	0.8333	0.9833	0.7778	0.0253	0.0225	0.9662	0.9704

4. Gemini Compact-Variant Experiment

The Gemini 2.5 Flash closed-model experiment used eight compact variants of the frozen final strategy. The run used temperature 0.0, max output tokens 64, seed 42 where supported, and thinking budget 0. The resulting CSV files are included under the results folders using the suffix compact_variants.

Main result: Gemini was strong in English, with one majority-vote error: the question "What does HTTP stand for?" tied between ABBREVIATION and DESCRIPTION and was resolved as DESCRIPTION. In Arabic, Gemini correctly handled NUMBER, LOCATION, PERSON, DESCRIPTION, and ENTITY, but all ten Arabic ABBREVIATION questions were classified as DESCRIPTION. This lowered Arabic accuracy to 0.8333 and macro-F1 to 0.7778.

5. Interpretation

The prompt progression confirms that explicit label definitions, few-shot examples, and a targeted DESCRIPTION/ENTITY rule reduce label-boundary confusion. Cross-model validation shows that the final prompt generalizes strongly across open models. Gemini's Arabic error pattern shows that a model can be stable across compact prompt variants while still being systematically wrong for a language-specific class.

6. Generated Files

results/raw_predictions/raw_predictions_crossmodel_gemini_2_5_flash_compact_variants.csv
results/sample_level/sample_level_results_crossmodel_gemini_2_5_flash_compact_variants.csv
results/metrics/metrics_crossmodel_gemini_2_5_flash_compact_variants.csv
results/class_level/class_level_results_crossmodel_gemini_2_5_flash_compact_variants.csv
results/error_analysis/error_analysis_crossmodel_gemini_2_5_flash_compact_variants.csv
results/prompt_level_errors/prompt_level_errors_crossmodel_gemini_2_5_flash_compact_variants.csv
results/summary_tables/bootstrap_ci_gemini_compact_variants.csv

7. Current Conclusion

The final prompt design is effective for the open models and remains strong for Gemini in English. The main unresolved issue is Arabic ABBREVIATION under the compact Gemini prompt variants. Future work should add more explicit Arabic acronym-expansion examples or a separate Arabic abbreviation rule.